SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	AB Dhanyashree	**Reg. No.:**	192472002
Section / Batch:	D/2024	**Date:**	28.07.26

Code of Conduct

I _AB Dhanyashree(192472002)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis •

Member 3: Demonstration, Case Study, Applications, and Conclusion •

GPS photo must submit.

- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04

Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10
Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

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(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Finite State Automaton (FSA) in Natural Language Processing

COURSE: NATURAL LANGUAGE PROCESSING
COURSE CODE: DSA0308

GUIDED BY:
Dr. KUMARAGURUBARAN. T

PRESENTED BY:
DHANYA SHREE AB(192472002)

Abstract

- Natural Language Processing (NLP) enables computers to understand and process human language.
- Finite State Automata (FSA) is one of the basic computational models used in NLP.
- It recognizes input patterns by moving through a sequence of predefined states.
- FSA is widely used for lexical analysis, tokenization, and regular expression matching.
- It offers a simple and efficient method for processing textual data.
- This presentation explains the concept, working principle, applications, advantages, and limitations of FSA in NLP.

Introduction

- A Finite State Automaton (FSA) is a mathematical model used to recognize patterns and sequences of characters.
- It consists of a finite number of states connected by transitions.
- The automaton starts from an initial state and processes one input symbol at a time.
- Based on the input, it moves between states according to predefined rules.
- If the automaton reaches a final (accepting) state after processing the input, the input is considered valid.
- FSA plays an important role in NLP applications such as lexical analysis, morphological parsing, and text pattern recognition.

Working, Components & Applications

Working of FSA

- The automaton starts from the initial state.
- It reads each input character sequentially.
- The transition function determines the next state.
- If the input reaches an accepting state, the string is accepted.
- If no valid transition exists, the string is rejected.
- The process continues until all input symbols are processed.

Main Components

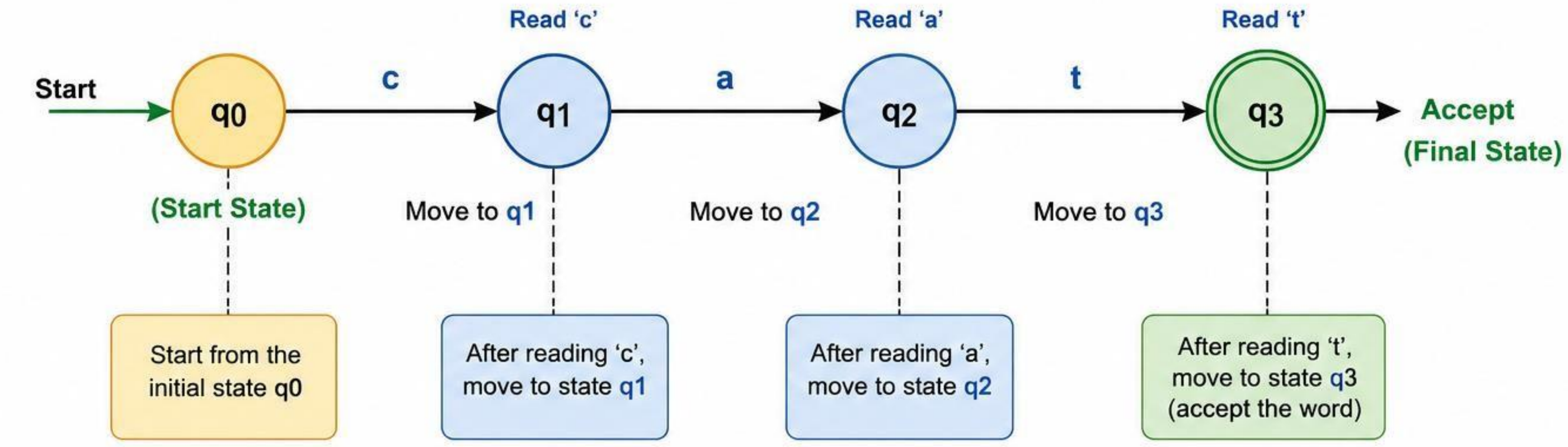
- States
- Input Alphabet
- Transition Function
- Start State
- Accept State

Applications

- Tokenization
- Spell Checking
- Regular Expression Matching
- Lexical Analysis
- Morphological Parsing
- Text Validation

FLOW DIAGRAM

EXAMPLE: FSA TO RECOGNIZE THE WORD "CAT"



TRANSITION TABLE

Current State	Input ('c')	Input ('a')	Input ('t')
q0	q1	-	-
q1	-	q2	-
q2	-	-	q3
q3	-	-	-

WORKING STEPS

1. FSA starts from the start state q0.
2. Read the first input symbol 'c'.
3. According to the transition function, move from q0 to q1.
4. Read the next input symbol 'a' and move from q1 to q2.
5. Read the next input symbol 't' and move from q2 to q3.
6. Since q3 is the final (accept) state, the input string "cat" is accepted.

Advantages & Disadvantages

Advantages

- Fast and efficient for pattern recognition.
- Simple to understand and implement.
- Requires very little memory.
- Provides accurate lexical analysis.
- Supports regular expression processing.
- Widely used in text processing applications.

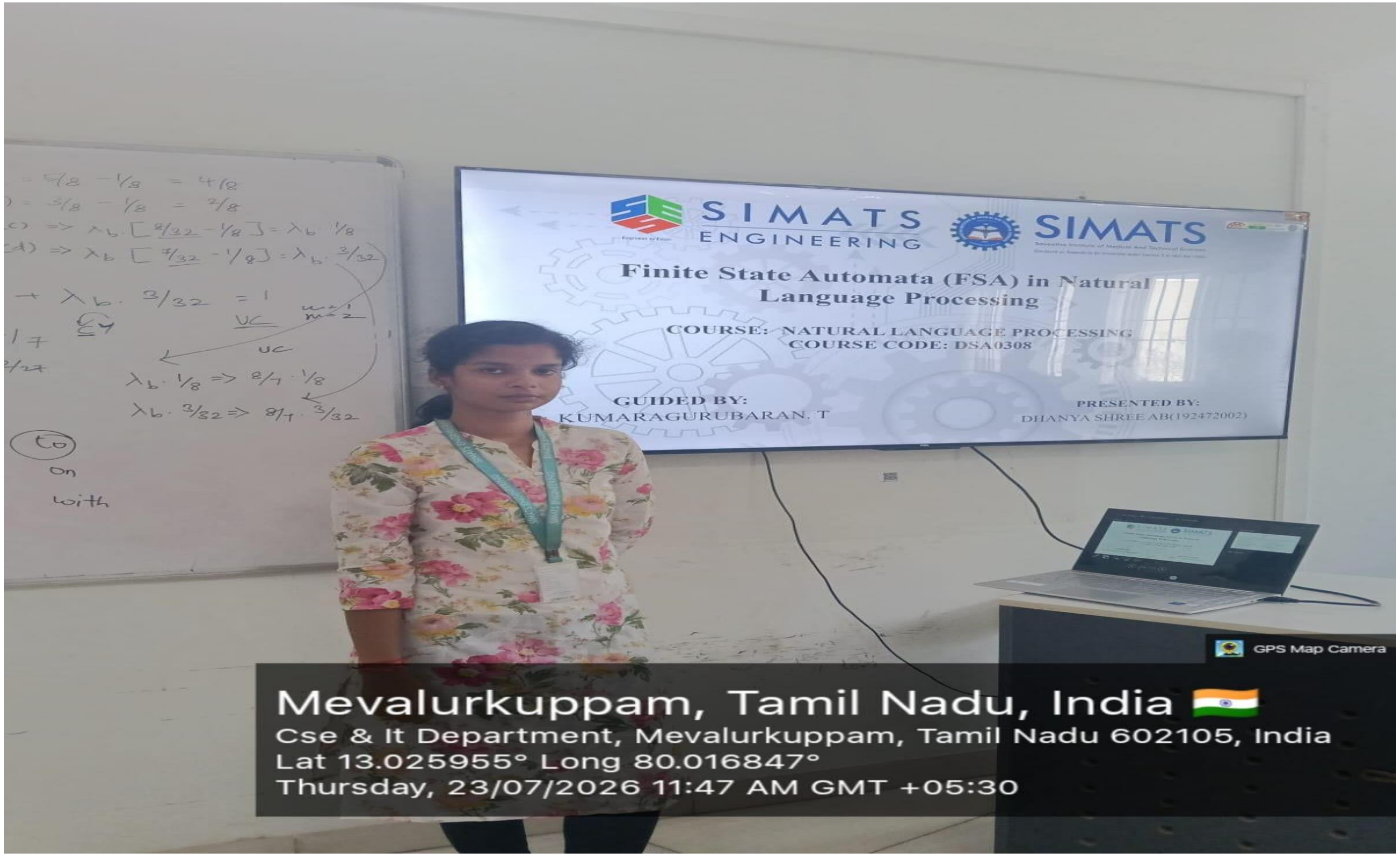
Disadvantages

- Cannot understand the meaning of words.
- Cannot process complex grammatical structures.
- Limited to finite memory.
- Does not handle context-dependent language.
- Unsuitable for semantic analysis.
- Less effective for complex NLP tasks.

Conclusion

- Finite State Automata is a fundamental model in Natural Language Processing.
- It recognizes words and patterns using states and transitions.
- FSA provides fast and efficient text processing.
- It is widely used in lexical analysis, spell checking, and tokenization.
- Although it has limitations, it remains an essential building block in NLP.
- Understanding FSA helps in learning advanced NLP concepts such as parsing and language modeling.

GEOTAG PHOTO



THANK YOU

Rubrics for Calculation

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	**Technical Content Accuracy**	**Problem Identification**	**Use of Tools**	**Communication**	**Professionalism**	**Total %**
2	**30**	**25**	**20**	**12**	**10**	**97**

Faculty In-charge	**Course Coordinator**	**Program Director**
KUMARAGURUBARAN T		
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________